

Requirements

Runtime Environment

Python ≥ 3.10

Dependencies

Package	Min Version	Purpose	Required
numpy	$\geq 1.24.0$	Numerical computation, random sampling	Yes
pandas	$\geq 2.0.0$	CSV loading and data processing	Yes
scikit-learn	$\geq 1.3.0$	RandomForestRegressor, MiniBatchKMeans	Yes
scipy	$\geq 1.11.0$	Mann–Whitney U statistical test, rankdata	Yes
optuna	$\geq 3.0.0$	Tree-structured Parzen Estimator (TPE comparison)	Optional
lightgbm	$\geq 4.0.0$	v3 LightGBM ensemble experiment	Optional
matplotlib	$\geq 3.7.0$	Report figure generation (generate_figures.py)	Optional

Installation

```
pip install -r requirements.txt
```

Per-Version Dependency Reference

Script	Required	Optional
flash_tuner.py (v1)	numpy, pandas, scikit-learn, scipy	—
flash_tuner_v2.py (v2)	numpy, pandas, scikit-learn, scipy	optuna (TPE)
flash_tuner_v3.py (v3)	numpy, pandas, scikit-learn, scipy, lightgbm	—
flash_tuner_v4.py (v4)	numpy, pandas, scikit-learn, scipy	—
flash_tuner_v5.py (v5)	numpy, pandas, scikit-learn, scipy	—
flash_tuner_v5_r30.py (v5-R30)	numpy, pandas, scikit-learn, scipy	—
analyze_v5_vs_v4.py	pandas	—
flash_tuner_v6.py (v6)	numpy, pandas, scikit-learn, scipy	—
flash_tuner_v7.py (v7)	numpy, pandas, scikit-learn, scipy	—
run_ablation.py	numpy, pandas, scikit-learn, scipy, optuna	—
run_extended_experiments.py	numpy, pandas, scikit-learn, scipy	optuna (TPE)
compare_budgets.py	numpy, pandas, scikit-learn, scipy	—
generate_figures.py	numpy, matplotlib	—

optuna is only required when running TPE comparison experiments. **lightgbm** is only required for the v3 experiment (macOS also requires `brew install libomp`). **matplotlib** is only required for generating report figures. The core tool v6 (**flash_tuner_v6.py**) has no optional dependencies.